

The Law of Large Numbers

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Introduction

The law of large numbers (LLN) is the formal statement of an intuition most students bring to statistics on day one: averages settle down as you collect more data. The precise statement comes in two flavours – the weak law and the strong law – and its consequences flow through estimation theory, simulation methodology, and machine learning.

Prerequisites

The reader should know what an expectation is, what a sample mean is, and should have seen `replicate()` or a similar simulation pattern in R.

Theory

Let $X_1, X_2, \dots$ be iid random variables with finite mean $\mu = E[X_1]$. The sample mean is $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$.

Weak law (WLLN): for every $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} P(|\bar{X}_n - \mu| > \varepsilon) = 0.$$

In words, the sample mean is eventually within any pre-specified distance of the population mean with probability approaching 1. This is convergence in probability: $\bar{X}_n \xrightarrow{P} \mu$.

Strong law (SLLN):

$$P\left(\lim_{n \rightarrow \infty} \bar{X}_n = \mu\right) = 1.$$

The sample mean converges to the population mean almost surely – for almost every realisation of the infinite sequence, the sample mean settles to μ . This is strictly stronger than the weak law.

Both laws depend on the existence of μ . For distributions without a finite mean (Cauchy, Pareto with tail index below 1), the sample mean does not converge at all; it remains wild regardless of sample size.

The LLN generalises beyond means: any continuous function of a sample mean is consistent for the same function of μ (continuous mapping theorem). This underwrites the plug-in principle and the consistency of most practical estimators.

Assumptions

The classical LLN requires iid observations with a finite mean. The LLN for non-iid sequences (stationary ergodic, martingale-difference, mixing) holds under appropriate generalisations but requires more care.

R Implementation

```

library(ggplot2)

set.seed(2026)
mu <- 5
sigma <- 3

n_vals <- seq(10, 10000, by = 50)

draws <- rnorm(max(n_vals), mean = mu, sd = sigma)
path <- data.frame(n = 1:length(draws),
                  running_mean = cumsum(draws) / seq_along(draws))

ggplot(path, aes(x = n, y = running_mean)) +
  geom_line(colour = "steelblue") +
  geom_hline(yintercept = mu, linetype = "dashed", colour = "red") +
  scale_x_log10() +
  labs(x = "n (log scale)",
       y = expression(bar(X)[n]),
       title = "LLN in action: running mean vs. population mean") +
  theme_minimal()

```

Plotting the running sample mean against n shows it converging to $\mu = 5$; different random seeds produce different paths, all converging to the same horizontal line.

Output & Results

The plot shows an initial ragged behaviour for small n , narrowing steadily as n grows, with the running mean nested closer and closer around the true 5. Classical theoretical rates: the typical deviation at sample size n scales as $\sigma/\sqrt{n}$ (the standard error).

Interpretation

The LLN justifies every Monte Carlo estimate and every use of the sample mean as a point estimate. A paper that reports “estimated by averaging over $N = 10^5$ simulation replicates” is implicitly invoking the LLN: the average converges to the population mean and, by the CLT, has a well-characterised standard error.

Practical Tips

- The LLN guarantees convergence but says nothing about speed – the CLT does. In practice, “large enough n ” depends on the population’s tail behaviour.
- Check that the mean exists before trusting the LLN. A single quiet assumption failure (sampling from a heavy-tailed population) turns a routine simulation into a misleading result.
- The LLN extends to sample variances, proportions, correlations, and more general functionals – any “mean of transformed data” is consistent.
- Running-mean plots are useful diagnostics in simulations: if the running mean is still drifting at the end of the simulation, more replicates are needed.
- In online learning and streaming algorithms, the LLN underlies the convergence of stochastic-gradient and recursive estimation schemes.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.

- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Scientific process and research workflow
- R, RStudio, Quarto, and renv toolchain
- Data types, tidy data, accuracy and precision
- Import, joins, and missingness with dplyr